

BELL 6G EXECUTIVE BRIEFING

15 minutes

Presented by



Making Bell's 6G Future Practical

Lower energy. Lower carbon. Safer autonomy.

How three PhD contributions translate into practical tools for Bell Mobility's 6G roadmap.

PRESENTER

Georges Parfait Djimefo Kapen

PhD candidate, Polytechnique Montréal — LARIM laboratory

THESIS DIRECTORS

Prof. Ranwa Al-Mallah · Prof. Samuel Pierre

DATE

Friday, May 15, 2026

For Bell Mobility executives

Bell already leads on 6G foundations. Four practical gaps remain.

Bell does not need another 6G demo. It needs working answers to four real operating constraints.

Energy

The network is getting more efficient, but Bell can't yet show how much each AI feature saves in electricity.

Carbon

Most of Bell's carbon footprint sits in purchased equipment and supplier activity — not only electricity.

Security

Future networks let software act on its own. Bell needs guardrails so attackers can't fool that software.

Canadian coverage

Rural, Indigenous, and Far-North service must be reliable on low power and limited backhaul.

THE THESIS OFFERS THREE PRACTICAL TOOLS

GAPF

Keep remote devices alive longer on tiny batteries.

CALASH

Choose network routes that save energy and delay equipment replacement.

CASTER-ZT

Let autonomous network software act fast, without trusting any input blindly.

Bell's 6G foundations are already in place

These six building blocks are what 6G will run on. Bell is already investing in every one of them.

1 5G Standalone & 5G+ Advanced

Bell positions 5G+ Advanced as the on-ramp to 6G — faster speeds, lower delay, network slicing.

2 AI inside the radio (with Ericsson)

Live tests showed up to ~20% higher download speed using AI inside Bell's network.

3 Cloud & Open RAN (with Nokia)

Radio software running on standard servers — the open architecture 6G will build on.

4 Edge computing & Bell AI Fabric

Bell's sovereign AI infrastructure on Canadian hydro-powered capacity.

5 Satellite-to-mobile (direct-to-cell)

Reaching the parts of Canada terrestrial towers cannot economically serve.

6 Climate commitments

BCE (Bell's parent company) targets ~58% lower direct emissions by 2030 — already 27% achieved.

What's missing in Bell's public 6G story today

Each gap is also a chance to show industry leadership before competitors do.

Per-feature energy reporting

Bell can report total electricity use, but not yet how much each AI feature in the network saves.

Why it matters: investor and regulator pressure on AI energy is rising.

A published rulebook for autonomous network software

Bell hasn't yet published a security model for software that controls the radio on its own (the part of Open RAN often called the "software brain").

Why it matters: Canadian cyber law (Bill C-8) is tightening.

No public pilots on emerging 6G hardware

New surface antennas, sub-THz radios, and sensing-radio (ISAC) ideas are research-stage today.

Why it matters: these are 3–5 year R&D bets, not deployment-ready.

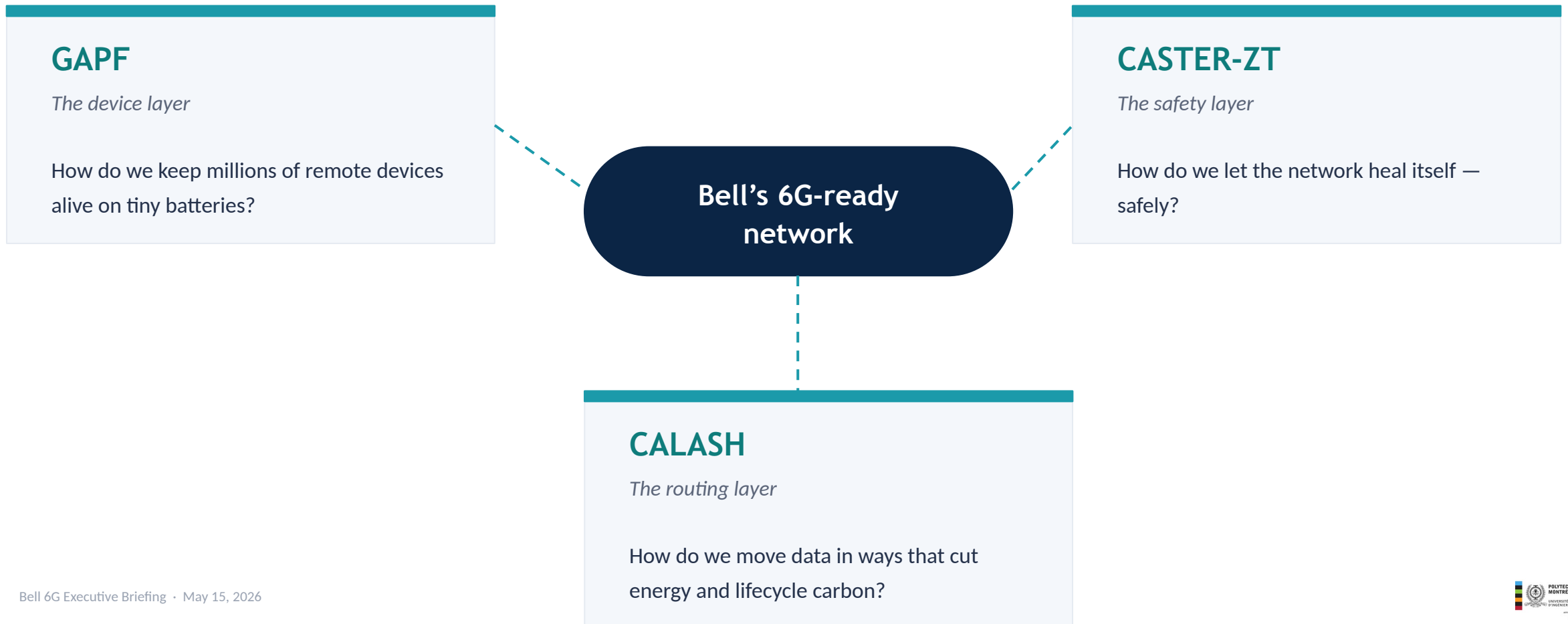
Rural, Indigenous, and public-safety constraints

Only ~66% of First Nations reserves have 50/10 Mbps service. CRTC now requires community consent for funded builds.

Why it matters: regulatory obligation and a brand opportunity.

Three control layers around Bell's 6G-ready network

Each layer answers one practical question Bell already asks today.



GAPF: keep remote devices alive longer on tiny batteries

THESIS RESULT

A tiny on-device routing brain (1,473 parameters) trained with multi-agent reinforcement learning.

98%

of messages delivered

2-3.5×

less energy used

13×

less memory needed

Sub-1 W

runs on a tiny IoT chip

What Bell can do with this

- **Longer-lasting field sensors**

Smart-meter, wildfire, flood, and farm sensors run years longer between battery swaps.

- **Fewer truck rolls**

Crews drive into the field only when actually needed — lower operating cost, lower fuel emissions.

- **Resilient public-safety meshes**

Devices keep coordinating locally even if a tower or backhaul link drops.

- **Massive IoT readiness**

Designed for the high device counts that 5G-Advanced and 6G will enable (“mMTC” = many low-power devices).

Bell teams that benefit: Bell IoT, Bell Mobility network ops, Public Safety

CALASH: choose routes that save energy and delay equipment replacement

THESIS RESULT

A routing controller that factors in three carbon sources at once: making the equipment, running it, and retiring it. Decisions adapt to how clean each provincial power grid is in real time.

+60%

longer equipment lifetime

Real-time

reacts to grid cleanliness

-33%

lower lifecycle carbon

ISO 14040

auditable methodology

What Bell can do with this

- **Address the bigger half of Bell's footprint**
Most of BCE's carbon impact sits in purchased equipment and supplier activity — not just electricity bills.
- **Use Canada's clean grids on purpose**
Shift work toward provinces with cleaner power when service-level rules allow.
- **Defer replacement carbon**
Routing that protects aging gear extends its useful life — reducing how often new hardware must be built and shipped.
- **Investor-defensible**
Standards-based (ISO 14040) so the savings story is auditable for CDP, regulators, and investors.

Bell teams that benefit: ESG / Sustainability, Network operations, Bell AI Fabric

CASTER-ZT: let autonomous network software act fast without trusting any input blindly

THESIS RESULT

A mathematically verified safety layer that screens decisions before they reach the live radio. It detects malicious or faulty inputs and blocks them — fast enough for real-time control.

74-98%

attackers detected

0

false alarms

~3 ms

primary test configuration

inside

Bell's real-time radio budget

What Bell can do with this

- **Catch rogue software inside the network**
Detects bad models, manipulated readings, or compromised modules — the threats Open RAN security working groups have prioritized.
- **Zero false alarms = no needless outages**
In a live network, a false alarm is itself a service outage. The model is tuned to never trigger one.
- **Fast enough for autonomous control**
~3 milliseconds in our primary test configuration — inside Bell's real-time radio budget.
- **Aligns with Canadian cyber law**
Maps to Bill C-8 telecom security obligations, supply-chain controls, and incident-reporting rules.

Bell teams that benefit: Bell Cyber, Bell Mobility RAN ops, Public Safety / PSBN

Bell can test all three tools today – on assets it already operates

Each tool maps onto a Bell environment that is already live. 6G standards will arrive later — the learning starts now.

Tool	Bell environment today	What we test	Why it future-proofs 6G
GAPF	Bell IoT / Smart Core sensor pilots	Run the controller on a small fleet of NB-IoT or LTE-M devices. Measure battery life and packet delivery in real Canadian conditions.	Carries into 6G's massive-IoT (many-device) use cases.
CALASH	Bell AI Fabric edge sites & Cloud RAN labs	Use real-time grid carbon data in routing decisions. Canadian provincial integration is the proposed pilot.	Carries into 6G energy and supply-chain reporting.
CASTER-ZT	Cloud / Open RAN sandbox with Nokia	Install the safety layer in a test radio controller. Inject attacker scenarios. Measure how fast and how cleanly it reacts.	Carries into 6G autonomous-recovery requirements.

Start small. Prove value. Then scale.

90 days

1

Prove the science

Independent code review, reproduce the results on hardware Bell trusts. Cost: small. Risk: low.

Small budget

6 months

2

One Bell pilot

Pick one of the three tools and one Bell scenario. Measure real KPIs against today's baseline.

Bounded scope

12-18 months

3

Scale & co-develop

Move what works into wider use. Map to Canadian cyber regulation, set IP and licensing terms.

Strategic commitment

AT EVERY STAGE BELL SEES:

Reproducible code · measurements on real hardware · side-by-side comparison to today's tools · clear path on IP and licensing

One workshop. One use case. One quarter to prove practical value.

Lead with the tool that closes the most visible gap first.

1 CALASH first CARBON & ENERGY

Closes Bell's most visible reporting gap. Carbon-aware routing using real-time grid carbon data. Canadian provincial integration is the proposed pilot.

ESG · Network operations · Bell AI Fabric

2 CASTER-ZT next AUTONOMY & SAFETY

Safety layer for autonomous radio software. A 3–5 year strategic collaboration with Bell Cyber.

Bell Cyber · Mobility · Public Safety

3 GAPF third RURAL & IoT

Tiny on-device routing brain for low-power and rural sensor fleets. Strong fit if Bell IoT prioritises long-life devices.

Bell IoT · Smart Core · Public Safety

We propose: one Bell-Polytechnique workshop. One Bell use case. One quarter to prove practical value.

THE TAKEAWAY

Bell can lead Canadian 6G by proving secure, carbon-aware autonomous networks — under Canadian constraints.

Three thesis contributions — GAPF, CALASH, CASTER-ZT — are ready to be tested against Bell's real workloads.

Thank you

Georges Parfait Djimefo Kapen · Polytechnique Montréal · LARIM

Directors: Prof. Ranwa Al-Mallah · Prof. Samuel Pierre

Bell 6G Executive Briefing · May 15, 2026

Source: [Bell 5G+ Advanced launch](#) · [Ericsson-Bell AI-native link adaptation](#) · [Nokia-Bell Cloud/Open RAN](#) · [BCE 2024 Climate Action Report](#) · [O-RAN Alliance security update 2025](#) · [CRTC Indigenous connectivity](#)



**POLYTECHNIQUE
MONTRÉAL**

UNIVERSITÉ
D'INGÉNIERIE

AFFILIATED WITH
Université
de Montréal

